

The empirical recurrence for OEIS A301783: recovering a conjecture that is not written down

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Abstract

OEIS A301783 records “Empirical recurrence of order 46”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 51$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 46, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 46 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A301783 is “Number of $n \times 7$ 0..1 arrays with every element equal to 0, 1, 2 or 3 horizontally or vertically adjacent elements, with upper left element zero.”. It has offset 1 and begins

64, 8192, 772272, 75632400, 7381982784, 720677122368, 70356843337600, 6868652110439488, ...

The entry states:

Empirical recurrence of order 46 (see link above).

As of the “Last modified” line on the live entry (Jun 23 2026 12:58:28, revision 6) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 16384$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 51$ of the 16384, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 51$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 102$ exact terms of the model returns order 46 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{46} c_i a(n-i),$$

c_1	81	c_{13}	−245002381751808	c_{25}	−1099998969726299537408	c_{37}	4243503056479
c_2	1880	c_{14}	182783226536192	c_{26}	−2635280112056574935040	c_{38}	1086369369798
c_3	−18844	c_{15}	9239346276636160	c_{27}	2019474999395844358144	c_{39}	−4205448796726
c_4	−624832	c_{16}	11671427035220992	c_{28}	6919679942278474891264	c_{40}	−2639740675476
c_5	572556	c_{17}	−240590263838529536	c_{29}	−2489100934658958295040	c_{41}	3257293000855
c_6	97098840	c_{18}	−684476064432861184	c_{30}	−13720363490380896796672	c_{42}	3585630802733
c_7	285283296	c_{19}	4202497477872095232	c_{31}	379758631217841307648	c_{43}	−1670778716399
c_8	−8881114592	c_{20}	17442391111501021184	c_{32}	18962518830240420069376	c_{44}	−2545575472058
c_9	−50490817536	c_{21}	−45921555626615226368	c_{33}	10965675113283426713600	c_{45}	4679023840055
c_{10}	506817660224	c_{22}	−219675845600641220608	c_{34}	−20677845633993222914048	c_{46}	8794197006764
c_{11}	4420797784960	c_{23}	326530108924509945856	c_{35}	−30557061382129437376512		
c_{12}	−16923479414272	c_{24}	882465674621091315712	c_{36}	17454909935762899206144		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 46, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $46 + 4$ terms removed returns order 46 as well, so $\deg q_{\min} = 46 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 12 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $46 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 46 that this sequence satisfies — and that family turns out to have exactly one member.

References

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